state-vectors, and from ordinary 3-vectors. However, because spin operators behave so much like 3-vectors, it does no harm to think of them in that way, and that's what we'll do here.

We measure spin components by orienting the apparatus $\mathcal{A}$ along any one of the three axes and then activating it. But then why not orient $\mathcal{A}$ along *any* axis and measure the component of σ along that axis? In other words, take any unit 3-vector $\hat{n}$ with components n_x, n_y, and n_z, and orient the apparatus $\mathcal{A}$ with its arrow along $\hat{n}$. Activating $\mathcal{A}$ would then measure the component of σ along the axis $\hat{n}$. There must be an operator that corresponds to this measurable quantity.

If σ really behaves like a 3-vector, then the component of σ along $\hat{n}$ is nothing but the ordinary dot product of σ and $\hat{n}$.[5,6] Let's denote that component of σ by σ_n, so that

$$\sigma_n = \vec{\sigma} \cdot \hat{n}$$

or, in expanded form,

$$\sigma_n = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z. \qquad (3.22)$$

To clarify the meaning of this equation, keep in mind that the components of $\hat{n}$ are just numbers. They themselves are not operators. Eq. 3.22 describes a vector-operator that is constructed as the sum of three terms, each containing a numerical coefficient n_x, n_y, or n_z. To be more concrete, we can write Eq. 3.22 in matrix form:

$$\sigma_n = n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Or even more explicitly, we can combine these three terms into a single matrix: